NEF ABUNDANCE: FRAMEWORK – FAILURE OF JET SEPARATION IN GENERAL DIMENSION

1. Failure of jet separation

Fix an integer $d \geq 2$. We use the following source inputs, denoted collectively by (I_d).

- For every projective klt $\mathbb{Q}$-pair (Y, B) of dimension $j < d$ and every nef rational Cartier divisor M, if $K_Y + B$ is pseudo-effective and, for some $a \in \mathbb{Q}_{>0}$,

$$(K_Y + B + aM) \cdot C > 0$$

  for every integral curve through a sufficiently very general point, then K_Y+B+cM is big for every $c \in \mathbb{Q}_{>0}$.
- Every projective log canonical $\mathbb{Q}$-pair of dimension at most d with nef adjoint has semiample adjoint.
- Every projective $\mathbb{Q}$-factorial generalized klt $\mathbb{Q}$-pair of dimension at most d, with fixed nef rational Cartier b-divisor data and pseudo-effective adjoint, admits a terminating choice of minimal-model program with a nef endpoint. The program extracts no divisors, preserves $\mathbb{Q}$-factorial generalized klt models, and on a common resolution the initial adjoint equals the final adjoint plus an effective divisor exceptional over the final model.

The first input is the lower-dimensional assertion (P_j) of [1, Section 3.1]. The last two are the companion inputs used in [1, Section 2]; here they are assumptions, not independently established higher-dimensional theorems.

Let X be a smooth projective variety over $\mathbb{C}$ of dimension d, with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor such that $\mathrm{n}(X, D) = d$ and $D^d = 0$. Assume (I_d).

Assume in addition that $\kappa(D) = -\infty$.

Setup.

- Fix an ample Cartier divisor A.
- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small. For integers $n \geq 1$, put

$$a_n = ((D + A/n)^d)^{1/d}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

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Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^d \to \epsilon^d$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \le a_n/2$ for all sufficiently large n. The symbol n indexes the sequence; the dimension is d.

- Let $p_i : X \times X \to X$ be the projections. Put

$$Z = \mathbb{P}_{X\times X}\big(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)\big), \qquad \pi : Z \to X \times X.$$

Let S_1, S_2 be the two summand sections and set $U = Z \setminus (S_1 \cup S_2)$. Thus $\dim Z = 2d + 1$.

- Set $\xi = c_1(\mathcal{O}_Z(1))$. For an integer $s > 0$, put

$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$

We use $\pi_*\mathcal{O}_Z(a) = \mathrm{Sym}^a(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D))$ for $a \ge 0$. Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with kP_n Cartier.

- Choose ϵ sufficiently small that

$$(14\epsilon)^d < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$

The integer s is chosen after ϵ and before n tends to infinity. Any jet-separating multiple is chosen after n.

Jet separation. Let W be smooth, let L be a Cartier divisor, and let $z \in W$. For $j \ge 0$, $|L|$ separates j-jets at z if

$$\forall u \in \mathcal{O}_{W,z}(L) \quad \exists \sigma \in H^0(W, \mathcal{O}_W(L)) \quad \sigma_z - u \in \mathfrak{m}_z^{j+1}\mathcal{O}_{W,z}(L).$$

Lemma 1.2. *For every positive integer s and every $n \gg 0$, $|kR_{n,s}|$ does not separate $(2d+2)k$-jets at any $z \in U$, for any positive integer k with kP_n Cartier.*

Draft status. This is a conditional, source-based proof framework following [1, Sections 4–6]. The source inputs (I_d) and the inherited moving-centre and Frobenius arguments have not been independently verified here. No unconditional higher-dimensional nonvanishing or abundance theorem is asserted.

2. Auxiliary estimates

The geometric argument will produce subvarieties with bounded degree and bounded canonical intersection. We first explain how these bounds contradict a growing lower bound for the degrees of curves. We then record the moving-center results from [12] that supply the required subvarieties. All varieties in this section are over $\mathbb{C}$.

2.1. **The bounded-degree obstruction.**

Lemma 2.1. *For every integer $f \geq 1$ there is an integer $m_f > 0$ with the following property. Let Y be a smooth integral projective variety of dimension f, let D_0 be a reduced simple normal crossing divisor or the zero divisor, and let L_0 be a nef integral Cartier divisor of full nef dimension on Y. Suppose that P_0 is a nef rational Cartier divisor and that r, C_0, C_1 are positive real numbers satisfying*

$$0 < P_0^f \leq C_0, \qquad P_0 - rL_0 \text{ is nef.}$$

If $A_0 = K_Y + D_0 + L_0$ is big and

$$A_0 \cdot P_0^{f-1} \leq C_1 P_0^f, \tag{1}$$

then

$$r \leq (m_f C_1)^{f-1} C_0. \tag{2}$$

The integer m_f depends only on f, and not on the variety or the divisors. For dimensions $1 \leq f \leq d$, one may replace all m_f by a single integer m depending only on d.

Proof. The pair (Y, D_0) is log canonical, its boundary coefficients belong to $\{0, 1\}$, and its nef polarizing divisor L_0 has Cartier index one. By [2, Theorem 1.3], an integer m_f depending only on these data makes $|m_f A_0|$ birational. Here A_0 is integral Cartier, so no rounding or additional denominator is involved.

Resolve the base ideal by a projective birational morphism $\mu : \widetilde{Y} \to Y$, with $\widetilde{Y}$ smooth, and write

$$\mu^*(m_f A_0) = H + F,$$

where H is the basepoint-free moving part and F is effective. The morphism defined by H is birational onto its image; in particular H is nef and big. In the rest of this proof, P_0 and L_0 denote their pullbacks to $\widetilde{Y}$. Effectivity of F and nefness of P_0 give

$$H \cdot P_0^{f-1} \leq m_f C_1 P_0^f. \tag{3}$$

For $f \geq 2$, put $s_i = P_0^{f-i} \cdot H^i$ for $0 \leq i \leq f$. Both divisors are nef and big, so the s_i are positive. The Khovanskii–Teissier inequalities [9, Corollary 1.6.3(i) and Example 1.6.4] $s_i^2 \geq s_{i-1} s_{i+1}$ show that the ratios s_i / s_{i-1} decrease. Hence

$$P_0 \cdot H^{f-1} \leq P_0^f \left(\frac{H \cdot P_0^{f-1}}{P_0^f} \right)^{f-1} \leq (m_f C_1)^{f-1} C_0. \tag{4}$$

Choose a very general point $y \in \widetilde{Y}$ over the isomorphism opens of both μ and the morphism defined by H. We also choose $\mu(y)$ in the very general locus that tests full nef dimension of L_0. Pull back $f - 1$ general hyperplanes through the image of y under

the morphism defined by H. This subsystem has only the base point y, so the general members meet properly and define an effective curve cycle of class H^{f-1}; the component C through y is a curve because their local equations are independent at y. Nefness gives

$$P_0 \cdot C \leq P_0 \cdot H^{f-1}.$$

The curve C is not contracted by μ, and its image passes through the chosen very general point. Full nef dimension and integrality therefore give $L_0 \cdot C \geq 1$. Since $P_0 - rL_0$ is nef, we obtain $P_0 \cdot C \geq r$, which proves (2).

If $f = 1$, take $C = Y$ directly. Then $P_0 \cdot Y \geq rL_0 \cdot Y \geq r$, and the required bound is $r \leq C_0$. Finally, [2, Theorem 1.3] applies to every multiple of its prescribed integer. A common multiple of the finitely many integers $m_1, \ldots, m_d$ therefore gives the last assertion. □

In particular, the hypotheses of Lemma 2.1 cannot hold along a sequence with $r \to \infty$ and with C_0, C_1 bounded, even if the varieties and divisors vary. The following observation explains how full nef dimension and pseudo-effectivity of the canonical divisor pass to the moving subvarieties used below.

Lemma 2.2. *Let X be a normal integral projective variety that is not uniruled, and let L be a nef Cartier divisor of full nef dimension on X. There is a complement X° of a countable union of proper closed subsets of X with the following property. If Y is a smooth integral projective variety and $g : Y \to X$ is generically finite onto a positive-dimensional image meeting X°, then g^*L has full nef dimension, Y is not uniruled, and K_Y is pseudo-effective.*

Proof. Choose X° so that every curve through any of its points has positive L-degree and no rational curve meets X°. The first condition is the very general characterization of full nef dimension. For the second, rational curves are parametrized by countably many algebraic families; the closure of the evaluation image of each such family is proper because X is not uniruled.

Since $g(Y)$ meets X°, it is contained in none of the closed sets excluded in its definition. A very general point of Y thus maps into X° and lies in the quasi-finite locus of g. Every curve through such a point has a curve as its image, so its g^*L-degree is positive. This proves full nef dimension. A covering family of rational curves on Y would similarly give a nonconstant rational curve meeting X°, a contradiction. The pseudo-effectivity of K_Y now follows from [3]. □

2.2. **Subadjunction and high-multiplicity components.** We use the following five results from the section *Tools for moving base components* of [12]. Their hypotheses distinguish generic information along a center from global information about the ambient

variety. Intersections on an integral subvariety are computed after pullback to its normalization and a smooth projective resolution. A base ideal is *isolated at the generic point* of a subvariety when its localization there is primary for the maximal ideal.

Lemma 2.3. *Let Z be a projective $\mathbb{Q}$-factorial klt variety, let $\Theta \geq 0$ be a rational divisor, and let $V \subset Z$ be integral of dimension $f > 0$. Suppose that (Z, Θ) is log canonical at the generic point of V and has a divisor of log discrepancy zero with center V. If P is a nef rational Cartier divisor on V and $\rho : V^* \to V$ is a smooth projective resolution factoring through its normalization, then*

$$K_{V^*} \cdot (\rho^* P)^{f-1} \leq (K_Z + \Theta)|_V \cdot P^{f-1}.$$

Only generic log canonicity is required. The proof in [12] uses the dlt modification of [6, Theorem 2.10], whose truncated boundary leaves an effective surplus supported over the non-lc locus. On a minimal stratum over V, adjunction gives a pair that is klt over the generic point of its connected-fiber base. This suffices for the klt-trivial canonical bundle formula and its b-nef moduli divisor [5, Definition 3.1 and Theorem 3.5]. The effective discriminant, the moduli divisor, and finite ramification have nonnegative intersections with the pulled-back nef class. The comparison uses canonical Weil cycles, so it requires no $\mathbb{Q}$-Gorenstein assumption on the normalization of V.

Lemma 2.4. *Let Z be a projective $\mathbb{Q}$-factorial klt variety of dimension e, let R be a nef rational Cartier divisor, and let $k > 0$ be an integer such that kR is Cartier. Let $0 \neq \mathcal{H} \subset H^0(Z, \mathcal{O}_Z(kR))$ be a linear subspace with proper base locus and base ideal $\mathfrak{b}$. Suppose that V is an integral base-locus component of dimension $f < e$, isolated at its generic point η_V, and that $\eta_V \in Z_{\mathrm{sm}}$. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{Z,\eta_V}$. If $\mathfrak{b}\mathcal{O}_{Z,\eta_V} \subset \mathfrak{m}^h$ for an integer $h > 0$, there are a rational number $0 < c \leq (e-f)k/h$ and an effective rational divisor $\Theta \sim_{\mathbb{Q}} cR$ such that (Z, Θ) is log canonical at η_V and has a log canonical place centered on V. The degree and canonical bounds are*

$$\begin{aligned} R^f \cdot V &\leq (k/h)^{e-f} R^e, \\ K_{V^*} \cdot R^{f-1} &\leq (K_Z + cR)|_V \cdot R^{f-1} \qquad (f > 0), \end{aligned} \tag{5}$$

where V^* is a smooth projective resolution. The degree bound also holds for $f = 0$.

The degree estimate is the local multiplicity bound h^{e-f} combined with successive general cuts by the given linear series. The boundary is obtained by averaging general members with total coefficient equal to the generic log canonical threshold of $\mathfrak{b}$. Thus it is an actual divisor to which Lemma 2.3 applies.

2.3. **Full jet kernels and restriction to a fiber.**

Lemma 2.5. *Let Z be an integral projective variety of dimension e, let R be a nef, big, semiample rational Cartier divisor, and fix an integer $k > 0$ such that kR is Cartier. Choose positive real numbers $\tau_0 < \cdots < \tau_e$ with consecutive gap $\delta > 0$. For $x \in Z_{\mathrm{sm}}$, put*

$$\mathcal{H}_i(x) = H^0\big(Z, \mathcal{O}_Z(kR) \otimes \mathfrak{m}_x^{\lceil k\tau_i \rceil}\big).$$

Suppose that for very general x all these spaces are nonzero and the base locus of $\mathcal{H}_0(x)$ has a positive-dimensional component through x. If $k\delta \geq 4$, then, after an algebraic parameter extension and restriction to nonempty opens, there are an integral parameter variety T with a dominant marked-point map $x : T \to Z_{\mathrm{sm}}$, a fixed index $i \in \{0, \ldots, e-1\}$, and a family of integral subvarieties V_t containing $x(t)$ and common to the base loci of $\mathcal{H}_i(x(t))$ and $\mathcal{H}_{i+1}(x(t))$. In particular, the incidence

$$\Gamma = \{(t, z) : z \in V_t\} \subset T \times Z$$

dominates Z.

For general t, the dimension f of V_t lies in $[1, e-1]$, V_t is an isolated component of the higher base locus at its generic point, and the entire higher-kernel base ideal is contained there in the ordinary power $\mathcal{I}_{V_t}^h$, where

$$h = \lceil k\delta/2 \rceil, \qquad k/h \leq 2/\delta.$$

The incidence base ideal is likewise contained in $\mathcal{I}_\Gamma^h$ on a dense open of Γ. These containments are taken in the smooth ambient open. If Z is also $\mathbb{Q}$-factorial and klt, then

$$\begin{aligned} R^f \cdot V_t &\leq (2/\delta)^{e-f} R^e, \\ K_{V_t^*} \cdot R^{f-1} &\leq (K_Z + cR)|_{V_t} \cdot R^{f-1}, \qquad 0 < c \leq 2(e-f)/\delta. \end{aligned} \tag{6}$$

The top intersection on V_t is positive when V_t meets the open on which the big semiample contraction is an isomorphism.

The use of the full kernels matters: differentiation in the parameter lowers vanishing at the moving mark and still gives a global section in the lower kernel. Incidence dominance makes these parameter directions span the normal directions. The vanishing of all derivatives of order less than h then gives the ordinary-power containment, which can be restricted to general fibers as follows.

Lemma 2.6. *In the setting of Lemma 2.5, let $g : Z \to B$ be a morphism. There is a dense open of the incidence with the following property. Choose (t, z) in this open, put $b = g(z)$, and suppose that V_t is smooth over a smooth open of its actual image at z. Let F be the reduced component through z of the scheme fiber of $V_t \to g(V_t)$. Near z in*

Z_b, the restricted higher-kernel base ideal has support exactly F and is contained in the ordinary power $\mathcal{I}_F^h$.

If F is proper in an integral ambient fiber component containing it, the restricted series is nonzero there and F is an isolated base component at its generic point. Applying Lemma 2.4 on that ambient component additionally requires its projectivity, $\mathbb{Q}$-factoriality, klt singularities, and smoothness at the generic point of F.

Here the fiber is chosen through a general incidence point. The statement does not assert the same conclusion for an arbitrary special fiber. Smoothness over the actual image identifies the restricted ideal of V_t with the reduced ideal of F locally; this is what preserves ordinary powers.

The $\mathbb{Q}$-factoriality needed for the base-component estimates can also be supplied by a small projective $\mathbb{Q}$-factorialization $\sigma : Z' \to Z$ of a klt ambient variety. This passage is made only for components that meet the isomorphism open of σ. Their strict transforms retain the generic base-ideal and multiplicity conditions for the pulled-back series, and the testing class $\sigma^* R$ is nef. Since $K_{Z'} = \sigma^* K_Z$, projection formula gives the same degree and canonical-intersection bounds on a common resolution. The pullback testing class need not be ample; nefness is the property used in these estimates. Passing to higher resolutions does not change the canonical intersections, because their additional exceptional divisors have zero intersection with the pulled-back testing class to the required power.

2.4. **Finite covers with a fixed branch complement.** A bound on the degree of a branched cover does not make the possible covers finite. The final companion input supplies the needed fixed branch complement from a family whose branch divisors do not sweep the base.

Lemma 2.7. *Let Y be a normal integral projective variety, let $\mathcal{V} \to T$ be a smooth projective family with integral fibers over an integral parameter variety, and let $g : \mathcal{V} \to Y$ be a morphism. Suppose that the general fiber maps $g_t : V_t \to Y$ are dominant and generically finite of degree at most a fixed integer e.*

After a finite parameter extension and shrinking, spread the geometric generic components of the zero divisor of the relative Jacobian over Y_{sm} to divisors $\mathcal{R}_1, \dots, \mathcal{R}_N$ in $\mathcal{V}$. Retain this notation after the base change. Let I consist of the indices j for which $\overline{g_t((\mathcal{R}_j)_t)}$ is a divisor in Y for general t. Assume that

$$\overline{g(\mathcal{R}_j)} \neq Y \qquad (j \in I).$$

After shrinking again, there is a smooth nonempty open $Y^\circ \subset Y$ over which every general finite normal Stein cover of g_t is finite étale. Only finitely many such normal covers of Y

occur, up to isomorphism over Y. The conclusion holds simultaneously for finitely many projections; the open and the finite lists may depend on all fixed parameters of the family.

To see the role of the hypothesis, remove Y_{sing} and the finitely many proper image closures $\overline{g(\mathcal{R}_j)}$ for $j \in I$. Every branch prime of a general Stein cover is accounted for by one of these relative Jacobian components. Purity therefore makes the restricted covers étale. The fundamental group of the resulting smooth complex variety is finitely generated, so it has only finitely many covers of degree at most e. Finally, a finite normal cover of Y is determined by its function field, hence by its restriction to this fixed open.

3. Geometric estimates

In the technical arguments below, put $L = D$, $t = 1/n$, $r = r_n$, $P = P_n$, and $q = s$. The letter L keeps the source notation for the fixed nef Cartier divisor; q is the bundle parameter.

$$P = r(L + tA), \qquad r \to \infty, \qquad P^d \to \epsilon^d. \tag{7}$$

For each fixed q, discard finitely many terms so that

$$\epsilon^d/2 \le P^d \le 2\epsilon^d, \qquad r > q + 1. \tag{8}$$

Every curve through a very general point of X has positive integral L-degree, hence P-degree at least r. Thus a covering family of curves of bounded P-degree will give a contradiction.

We specify how the same obstruction applies to subvarieties and covers. A resolution $W \to X$ has an effective canonical divisor: pull back a nowhere-zero canonical form on the smooth locus of X and use terminality. Hence W, and therefore X, is not uniruled [3]. By Lemma 2.2, if a smooth projective variety Y maps generically finitely to a positive-dimensional subvariety containing a sufficiently very general point of X, then Y is not uniruled and the pulled-back Cartier divisor L_Y has full nef dimension. In particular, K_Y is pseudo-effective. If $f = \dim Y < d$, the induction hypothesis (P_f) applied to $(Y, 0)$ gives

$$K_Y + L_Y \text{ big}. \tag{9}$$

All occurrences of "very general" below include the fixed conditions of Lemma 2.2.

Consequently, for $f < d$ it suffices to find on such a Y a nef testing divisor P_Y with

$$0 < P_Y^f \le C_0, \qquad P_Y - rL_Y \text{ nef}, \qquad K_Y P_Y^{f-1} \le C P_Y^f,$$

where C_0, C are independent of t. Indeed $L_Y P_Y^{f-1} \le P_Y^f/r$, so Lemma 2.1 applies. We will also use that lemma in dimension d, after proving the required bigness separately.

For the rest of this section, a bound is called uniform if it is independent of t; it may depend on parameters fixed before t tends to zero. Since the dimensions lie in a finite range, Lemma 2.1 excludes all such configurations once r is sufficiently large.

3.1. **The scalar jet estimate.** Put $\rho = (P^d)^{1/d}$. The first estimate bounds the vanishing order of every section at a very general point; it is stronger than a bound on a single fixed linear system.

Lemma 3.1. *In the setting of (7), for all sufficiently small t and at a very general point $x \in X$, every nonzero section of kP, where k is a positive integer and kP is Cartier, satisfies*

$$\operatorname{ord}_x(s) \le 4k\rho. \tag{10}$$

Proof. Suppose that violations occur for arbitrarily small t. Fix such a t. At a very general point a violation for some k implies that the corresponding complete jet kernel is nonzero at the general point. Indeed the kernel ranks are algebraic rank conditions on the smooth locus, and there are only countably many choices of k and of the required integral order. Raising the violating section to powers allows k to be arbitrarily large and divisible.

Choose $d+1$ levels

$$\tau_i = \frac{3}{2}\rho + i\delta \quad (0 \le i \le d), \qquad \delta = \frac{2\rho}{d}.$$

They all lie strictly between ρ and 4ρ. For a sufficiently large divisible k, their full jet kernels are nonzero and $k\delta \ge 4$. The lowest kernel has a positive-dimensional base component through its marked point. Otherwise that point would be an isolated base component of multiplicity at least $h_0 = \lceil k\tau_0 \rceil$. The zero-dimensional case of Lemma 2.4 would give

$$1 \le (k/h_0)^n P^d < 1,$$

a contradiction.

Lemma 2.5 now supplies a moving family of proper base components V, with dominant incidence and dimension $1 \le f < d$. On a resolution V^*, write P_V, L_V for the restrictions followed by pullback. The lemma gives

$$0 < P_V^f \le (2/\delta)^{d-f} P^d, \qquad K_{V^*} P_V^{f-1} \le c P_V^f, \qquad 0 < c \le \frac{2(d-f)}{\delta}.$$

Here the canonical term of the ambient variety vanishes because $K_X \sim 0$. The volume window (8) bounds δ above and away from zero, so these are uniform estimates. The incidence point may be chosen over a very general point of X. Thus (9) gives bigness of $K_{V^*} + L_V$, and $P_V - rL_V$ is nef. Lemma 2.1 contradicts $r \to \infty$. □

4. The fixed witness

Lemma 4.1. *For every fixed $s > 0$ and every $n \gg 0$, there is a very general point $x \in X$ such that, on its blowup $\beta : \widetilde{X} \to X$ with exceptional divisor E, there are a birational morphism $f : V \to \widetilde{X}$ from a smooth projective variety and ample Cartier divisors $H_1, \dots, H_{d-1}$ on V satisfying*

$$f^*\beta^*(P_n + sD) \cdot H_1 \cdots H_{d-1} < 40\epsilon\, f^*E \cdot H_1 \cdots H_{d-1}.$$

Proof. Put $\rho_n = (P_n^d)^{1/d}$ and choose n sufficiently large that Lemma 3.1 applies, $r_n > s$, and $\rho_n < 2\epsilon$. Every nonzero section of $2kP_n$ then has order at most $8k\rho_n < 16k\epsilon$ at a very general point x. The divisor

$$\beta^*(2P_n) - 40\epsilon E$$

is not pseudo-effective. Otherwise its convex combination with the big divisor $\beta^*(2P_n)$ would make $\beta^*(2P_n) - cE$ big for some rational $16\epsilon < c < 40\epsilon$. A sufficiently divisible multiple would have a section of order at least kc, a contradiction.

By pseudo-effective/strongly-movable duality [3], choose one actual generating complete intersection on a smooth birational model whose pairing with this divisor is negative. Approximate the ample real classes by rational ones and clear denominators to obtain $f, H_1, \dots, H_{d-1}$ with

$$f^*\beta^*(2P_n) \cdot H_1 \cdots H_{d-1} < 40\epsilon\, f^*E \cdot H_1 \cdots H_{d-1}.$$

Since $P_n - sD$ is nef, the required inequality follows. The denominator is positive because f^*E is nonzero effective and all the H_i are ample. Fix this model and these divisors before reduction. □

Lemma 4.2. *For every ample rational Cartier divisor P on X, there are positive integers $l_1, \dots, l_{d-1}$ and a smooth integral complete-intersection flag of successive classes l_iP, ending in a curve C, such that*

$$\mu_{\min}(\Omega^1_X|_C) \geq 0.$$

Proof. Since $K_X \sim 0$, X is not uniruled. Generic semipositivity of Ω^1_X [4, Theorem 2.1] and restriction of its Harder–Narasimhan filtration to sufficiently general complete intersections [10, Theorem 6.1] give the asserted slope bound. Choose the l_i sufficiently large and divisible that l_iP is very ample; Bertini gives the smooth integral flag. □

5. The Frobenius obstruction

The finite-data statement below is [1, Theorem 6.1], specializing the companion Frobenius comparison. Its divisor G is a rational polarization and is distinct from the fixed

nef Cartier divisor D in the Setup. All models, sections, the flag, and the complete-intersection witness are fixed before reduction.

5.1. **The finite-data statement.** For a vector bundle on a smooth projective curve, its slope is degree divided by rank. The least and greatest Harder–Narasimhan slopes are denoted by $\mu_{\min}$ and $\mu_{\max}$. For a line bundle on a smooth variety, to generate jets through order a at a point means surjectivity to its stalk modulo the $(a+1)$-st power of the maximal ideal.

Proposition 5.1. *Let W be a smooth connected projective complex variety of dimension $d \geq 2$. Let G, H be rational Cartier divisors, with H ample, let S be an integral Cartier divisor, and fix an integer $q > 0$ and real numbers $r > 1$, $\epsilon > 0$. Suppose*

$$H^d \leq (2\epsilon)^d, \qquad K_W H^{d-1} \leq 2H^d/r, \qquad (2G + qS)H^{d-1} \leq 4H^d, \tag{11}$$
$$(14\epsilon)^d < \tfrac{1}{4}, \qquad 80\epsilon < \tfrac{3}{4}.$$

Assume that the following fixed data exist.

(i) *A smooth integral complete-intersection flag from W to a curve C, whose successive divisor classes are $h_i = l_i H$ for fixed positive integers l_i for which $l_i H$ is Cartier, with $\mu_{\min}(\Omega^1_W|_C) \geq 0$.*

(ii) *On the projective bundle*

$$Z = \mathbb{P}\big(\mathcal{O}_W(S)_1 \oplus \mathcal{O}_W(S)_2\big) \longrightarrow W \times W, \qquad \xi = c_1(\mathcal{O}_Z(1)), \qquad M = G_1 + G_2 + q\xi,$$

use the convention that the direct image of $\mathcal{O}_Z(a\xi)$ is the a-th symmetric power of the displayed sum for $a \geq 0$; subscripts indicate pullback from the corresponding factor of $W \times W$. The two axes are the sections defined by the two summands. A smooth point z_* in their complement and finitely many sections of $k_0 M$, for some integer $k_0 > 0$ with $k_0 G$ Cartier, generate jets through order $(2d+2)k_0$ at z_*.

(iii) *A point $x \in W$, its blowup $\pi_x : \widehat{W} \to W$ with exceptional divisor J, and a curve class*

$$\gamma = f_*(A_1 \cdots A_{d-1}),$$

where $f : V \to \widehat{W}$ is one fixed birational morphism, V is smooth integral projective, and the A_i are ample integral Cartier divisors, such that $J \cdot \gamma > 0$ and

$$\pi_x^*(G + K_W/2 + \lambda S) \cdot \gamma \leq 40\epsilon J \cdot \gamma \qquad (0 \leq \lambda \leq q). \tag{12}$$

It suffices to check the two endpoints in this affine inequality.

These data cannot coexist.

The point x is independent of the two base coordinates of z_*. No nefness is required of G, S, or K_W, and the theorem does not assume that K_W is pseudo-effective. Its hypotheses specify the flag and the movable curve class themselves. All these data will be fixed before the residual characteristic tends to infinity.

5.2. **Reduction and Frobenius jets.** We begin the proof of Proposition 5.1 by spreading its hypotheses and converting the fixed ordinary jets into jets of order proportional to the residual characteristic. Put $\Lambda = \prod_{i=1}^{d-1} l_i$. The flag satisfies

$$[C] = \Lambda H^{d-1}, \qquad h_i \cdot C = l_i \Lambda H^d. \tag{13}$$

Choose one sufficiently ample integral Cartier divisor T_0 so that every $T_0 + aS$, for integers $0 \le a \le (k_0 - 1)q$, has a section nonvanishing at each of the two base coordinates of z_*. Fix these finitely many sections. All models, morphisms, divisors, points, the flag, the complete-intersection witness for γ, and the jet and filler sections spread over an integral finitely generated $\mathbb{Z}$-algebra of characteristic zero. Shrink its spectrum so that the fibers and flag are smooth projective and geometrically integral, the birational morphisms remain birational, the fixed ample bundles remain ample, and the finite jet surjection and nonvanishings persist. Birationality is preserved by spreading an isomorphism between dense opens. All displayed intersection numbers are constant.

We also preserve $\mu_{\min}(\Omega^1_W|_C) \ge 0$. One way is to spread its characteristic-zero Harder–Narasimhan filtration, make all its factors locally free, and use openness of their semistability on curves; their degrees and ranks are fixed. Equivalently, a single relative ample twist globally generates the curve bundle. Every quotient of rank a then has degree bounded below by $-ad$ for a fixed integer e. A negative-degree quotient therefore has one of finitely many ranks and degrees. The proper relative Quot schemes for those Hilbert polynomials have images missing the generic point: after removal of torsion, a negative-degree coherent quotient there would give a negative-degree vector-bundle quotient. Removing those finitely many images proves the same openness assertion directly.

This base has closed points in arbitrarily large characteristics. Take algebraic closures of their residue fields and exclude denominator primes. Retain the notation for the reductions. The curve γ continues to pair nonnegatively with every effective divisor: pull that divisor back under the fixed birational morphism and intersect with its fixed ample divisors. This tests effective divisors on the reduction itself. No specialization of a characteristic-zero effective cone is used.

Lemma 5.2. *For a sufficiently large residual characteristic p, set*

$$N_p = \lfloor p/k_0 \rfloor, \qquad B_p = k_0 N_p D + T_0.$$

The sections of $pq\xi + B_{p,1} + B_{p,2}$ generate jets through order $(2d+2)k_0N_p$ at z_*. In particular, they surject onto the quotient of its local ring by the p-th powers of all regular parameters.

Proof. Multiply N_p copies of the fixed jet system. Every monomial of total degree at most $(2d+2)k_0N_p$ is a product of N_p monomials of degree at most $(2d+2)k_0$. Taking sections with those leading monomials gives a triangular system, ordered by total degree. Starting at degree zero and removing the error in each successive degree proves the required jet surjection. This argument takes place in the local ring and never divides by factorials.

Put $e_p = (p - k_0N_p)q$, so $0 \le e_p \le (k_0-1)q$. Among the sections fixed before reduction, choose

$$u_0 \in H^0(W, \mathcal{O}_W(T_0)), \qquad v_{e_p} \in H^0(W, \mathcal{O}_W(T_0 + e_pS))$$

nonzero at the first and second base coordinates of z_*, respectively. Their tensor product belongs to the summand of first-slot weight zero in the projective-bundle formula; its fiber monomial is the e_p-th power of the second coordinate. It defines a section of $e_p\xi + T_{0,1} + T_{0,2}$ nonzero at z_*, since z_* lies off both axes. Multiplying the product jet system by this section changes its divisor from N_pk_0M to $pq\xi + B_{p,1} + B_{p,2}$. Multiplication is invertible on the local jet algebra. Only the finitely many previously fixed values of e_p occur.

Since $\dim Z = 2d+1$, the quotient by the p-th powers of regular parameters has top total degree $(2d+1)(p-1)$. For large p this is at most $(2d+2)k_0N_p$. The underlying ordinary/Frobenius ideal comparison is also recorded in [11, proof of Proposition 2.12, Equation (2.8)]. □

5.3. **Full rank from the two-slot jets.** Let $F : W \to W'$ be relative Frobenius to the base-field twist. It is finite flat because W is smooth. Write S' for the twist of S, so $F^*S' = pS$, and define

$$\mathcal{E} = F_*\mathcal{O}_W(B_p), \qquad s = \operatorname{rk}\mathcal{E} = p^d, \qquad R = s^2.$$

For integers $0 \le a \le q$, put $U_a = H^0(W, \mathcal{O}_W(B_p + paS))$. Projection formula gives an evaluation map on $W' \times W'$:

$$\bigoplus_{\substack{a+b=q\\ a,b\ge 0}} U_a \otimes U_b \otimes \mathcal{O}(-aS'_1 - bS'_2) \longrightarrow \mathcal{E} \boxtimes \mathcal{E}. \tag{14}$$

Lemma 5.3. *The map in (14) has generic rank R.*

Proof. Trivialize the two summands defining Z near the selected base points. Let z be the torus coordinate, whose value at z_* is $z_0 \neq 0$. By the projective-bundle formula, the sections in Lemma 5.2 are weight polynomials in z with weight-j coefficients in

$$H^0(W, B_p + jS) \otimes H^0(W, B_p + (pq-j)S), \qquad 0 \le j \le pq.$$

Let A_* be the tensor product of the local algebras of the two base Frobenius fibers. The local quotient in Lemma 5.2 is

$$A_*[z]/(z^p - z_0^p),$$

free over A_* with basis $1, z, \dots, z^{p-1}$. Project to the coefficient of 1 in this basis. Exactly the weights $j = pa$ survive, with multipliers z_0^{pa}. Their coefficients are the values of the corresponding summands of (14), in the chosen frames and the induced twisted frames. As the full jet map is surjective, these coefficients span A_*, of dimension $p^{2d} = R$. Thus the evaluation has full rank at this pair of base points, and hence generically. The calculation uses a basis over the possibly nonreduced algebra A_*, so reducedness of the Frobenius fibers is unnecessary. □

The jet calculation at z_* proves generic full rank; the diagonal estimate below will be tested at (x', x'), obtained from the separately fixed point x.

5.4. **Sections on the Frobenius diagonal.** The Frobenius fiber product has the cartesian square

$$\begin{array}{ccc} \Delta_F = W \times_{W'} W & \xrightarrow{\mathrm{pr}_2} & W \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle F} \\ W & \xrightarrow[F]{} & W'. \end{array} \tag{15}$$

Its reduced subscheme is the diagonal W. Put

$$\mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + pqS_1).$$

Write $h = F \circ \mathrm{pr}_1 = F \circ \mathrm{pr}_2 : \Delta_F \to W'$. On this fiber product,

$$\mathcal{O}(pS_1) \simeq h^*\mathcal{O}_{W'}(S') \simeq \mathcal{O}(pS_2).$$

Consequently every pair of weights $a + b = q$ gives the same line bundle:

$$\mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + paS_1 + pbS_2) \simeq \mathcal{L}_p.$$

On the diagonal of $W' \times W'$, every source twist of (14) becomes $-qS'$. Finite base change and projection formula identify the twisted target as

$$(\mathcal{E} \otimes \mathcal{E})(qS') \simeq h_*\mathcal{L}_p.$$

The product sections defining the columns restrict to global sections of this one bundle $\mathcal{L}_p$. Their values at a diagonal point therefore lie in the image of $H^0(\Delta_F, \mathcal{L}_p)$ in the corresponding fiber of $h_*\mathcal{L}_p$. This proves that the rank at every diagonal point is at most

$$h^0(\Delta_F, \mathcal{L}_p). \tag{16}$$

For a rank-d bundle V in characteristic p, denote by $A_j(V)$ the degree-j part of its symmetric algebra modulo the p-th powers of local generators. This construction is independent of frame: the p-th power of a linear combination is the sum of the p-th powers in characteristic p. Each $A_j(V)$ is locally free and a quotient of $V^{\otimes j}$. Set

$$u = d(p-1), \qquad e_j = \operatorname{rk} A_j(V), \qquad \sum_{j=0}^{u} e_j = p^d.$$

Filtering $\mathcal{L}_p$ by powers of the ideal of the reduced diagonal in Δ_F gives the graded bundles

$$\mathcal{G}_j = \mathcal{O}_W(2B_p + pqS) \otimes A_j(\Omega^1_W), \qquad 0 \le j \le u. \tag{17}$$

Indeed, in smooth local coordinates the algebra is generated by parameter differences with their p-th powers zero, and its degree-one conormal is Ω^1_W. This coordinate calculation can be made étale locally or in completions and identifies the global associated graded. This is the diagonal-ideal form of the canonical Frobenius filtration [7, Corollary 3.5]. The same calculation, using only the bundle from the second factor, filters $F^*\mathcal{E}$ with pieces [13, Theorem 3.7]

$$\mathcal{O}_W(B_p) \otimes A_j(\Omega^1_W). \tag{18}$$

Complementary multiplication is a perfect pairing, giving

$$A_{u-j}(V) \simeq A_j(V)^\vee \otimes (\det V)^{p-1}. \tag{19}$$

On monomials, each exponent vector pairs with its complement to $(p-1,\dots,p-1)$. The top line transforms by the character $(\det V)^{p-1}$: this is immediate on diagonal matrices and hence identifies the character of GL_d on that line.

5.5. **A uniform bound for the diagonal rank.** We bound the sections of every graded bundle in (17) by its rank times the same scalar polynomial in p. This uniformity lets us sum the ranks of the graded pieces without an additional factor for their number. We begin with a slope bound on the fixed curve; here slope means ordinary degree divided by rank.

Lemma 5.4. *There is a constant $c \ge 0$, independent of the sufficiently large residual characteristic p and of j, such that for $V = \Omega^1_W|_C$ one has*

$$\mu_{\max}(A_j(V)) \le (p-1)K_W \cdot C + nc.$$

Proof. Write F_C for absolute Frobenius of C. Langer's instability estimate [8, Corollary 2.5] gives

$$L_{\min}(V) := \lim_{e\to\infty} p^{-e}\mu_{\min}(F_C^{e*}V) \ge -\frac{c}{p-1}. \tag{20}$$

To make the uniformity explicit, let g be the fixed genus of C. One can choose a nef line bundle A_C of degree at most $e_g = \max\{4g-2, 0\}$ such that $T_C \otimes A_C$ is globally generated; for $g \leq 1$, take $A_C = \mathcal{O}_C$. Langer's bound is

$$\mu_{\min}(V) - L_{\min}(V) \leq \frac{(\operatorname{rk} V - 1)\deg A_C}{p-1}.$$

Since $\operatorname{rk} V = d$ and $\mu_{\min}(V) \geq 0$, we may take $c = (d-1)e_g$, independently of the reduction.

We also need, for every integer $i \geq 0$,

$$\mu_{\min}(V^{\otimes i}) \geq iL_{\min}(V). \tag{21}$$

Fix p and i. Choose a line bundle A_e of degree $-\mu_{\min}(F_C^{e*}V) + O(1)$, rounded up with a fixed genus-dependent additive constant, so that $F_C^{e*}V \otimes A_e$ is globally generated. This follows from Serre duality: after subtracting any point, its minimum slope can be made greater than $2g(C) - 2$, which annihilates H^1 and makes evaluation at that point surjective. For every vector-bundle quotient Q of $V^{\otimes i}$, the bundle $F_C^{e*}Q \otimes A_e^{\otimes i}$ is then globally generated. Consequently

$$p^e\mu(Q) + i\deg A_e \geq 0.$$

Divide by p^e and let $e \to \infty$ to prove (21). This limit is taken for each fixed p, i; there is no interchange of Frobenius-iteration and characteristic limits.

Since $A_{u-j}(V)$ is a quotient of $V^{\otimes(u-j)}$, the perfect pairing (19) yields

$$\begin{aligned}\mu_{\max}(A_j(V)) &= (p-1)\deg\det V - \mu_{\min}(A_{u-j}(V)) \\ &\leq (p-1)K_W \cdot C + \frac{(u-j)c}{p-1} \\ &\leq (p-1)K_W \cdot C + nc.\end{aligned}$$

□

The divisors $B_p - pD = T_0 - (p - k_0N_p)G$ range over a fixed finite list. Combining (11), (13), and Lemma 5.4 therefore gives, uniformly for $0 \leq j \leq u$,

$$\begin{aligned}\mu_{\max}(\mathcal{G}_j|_C) &\leq p(4 + 2/r)\Lambda H^d + O(1) \\ &\leq M_p := 7p\Lambda H^d + c_0,\end{aligned} \tag{22}$$

where $c_0 \geq 0$ is fixed. The coefficient 7 provides harmless room in this common bound.

Lemma 5.5. *For all sufficiently large p,*

$$h^0(\Delta_F, \mathcal{L}_p) \leq R\big(7^nH^d + O(1/p)\big) < R/4.$$

The error constant is independent of the filtration index j.

Proof. Write the fixed flag as

$$W = W_d \supset W_{d-1} \supset \cdots \supset W_1 = C, \qquad W_{d-k} \in |h_k|_{W_{d-k+1}} \quad (1 \le k \le d-1).$$

At step k, fix the preceding nonnegative integers $b_1, \dots, b_{k-1}$ and put

$$\mathcal{F}_k = \mathcal{G}_j|_{W_{d-k+1}}\Big(-\sum_{i<k} b_i h_i\Big).$$

For each $b_k \ge 0$, the divisor restriction sequence is

$$0 \longrightarrow \mathcal{F}_k(-(b_k+1)h_k) \longrightarrow \mathcal{F}_k(-b_k h_k) \longrightarrow \mathcal{F}_k(-b_k h_k)|_{W_{d-k}} \longrightarrow 0.$$

Iterating it for $b_k = 0, 1, \dots$ and then proceeding to the next member of the flag bounds the section space by

$$h^0(W, \mathcal{G}_j) \le \sum_{b_1,\dots,b_{d-1}\ge 0} h^0\left(C, \mathcal{G}_j|_C\Big(-\sum_i b_i h_i|_C\Big)\right). \tag{23}$$

At each stage the remainder is zero after sufficiently negative ample twisting. Its stopping index need not be uniform in j, p or in the preceding twists: enlarging the nonnegative finite sums to the displayed lattice sum preserves the inequality.

A summand in (23) vanishes when $\sum_i b_i(h_i \cdot C) > M_p$, because then its maximum slope is negative. Each nonzero summand has at most $e_j(M_p+1)$ sections. Indeed, evaluation at $\lfloor M_p \rfloor + 1$ distinct points is injective; the kernel has negative maximum slope after subtracting those points. By (13) it follows that

$$\begin{aligned} h^0(W, \mathcal{G}_j) &\le e_j(M_p+1)\prod_{i=1}^{d-1}\left(1 + \frac{M_p}{l_i \Lambda H^d}\right) \\ &= e_j p^d\big(7^n H^d + O(1/p)\big). \end{aligned} \tag{24}$$

The leading coefficient follows from $\Lambda = \prod_i l_i$. More importantly, the whole scalar polynomial on the first line is the same for every j.

Sum over the filtration in (17). The ranks satisfy $\sum_j e_j = p^d$, so (24) gives the asserted bound with $R = p^{2d}$. There is no extra factor for the number of graded pieces. Finally

$$7^n H^d \le (14\epsilon)^d < 1/4$$

by (11), proving the strict inequality for all large p. □

5.6. **The two orders of the determinant.** Choose R generically independent columns of (14) from bases of its source vector spaces. Their determinant is a nonzero section

$$0 \ne \sigma \in H^0(W' \times W', \mathcal{Q}_1 \boxtimes \mathcal{Q}_2)$$

of an actual external-product line bundle. To specify its factors, let m_i be the sum of the weights of the R chosen columns in slot i. The target determinant is $(\det \mathcal{E})^s \boxtimes (\det \mathcal{E})^s$, and the chosen source determinant is $\mathcal{O}(-m_1 S') \boxtimes \mathcal{O}(-m_2 S')$. Thus

$$\mathcal{Q}_i = (\det \mathcal{E})^{\otimes s} \otimes \mathcal{O}_{W'}(m_i S'), \qquad \lambda_i = m_i/R.$$

Every column weight lies between 0 and q, so $0 \le \lambda_i \le q$ independently of the columns chosen.

Under the numerical identification with the base-field twist,

$$\frac{c_1(\mathcal{Q}_i)}{R} = \frac{c_1(\mathcal{E})}{s} + \lambda_i S = \frac{B_p}{p} + \frac{p-1}{2p} K_W + \lambda_i S, \qquad 0 \le \lambda_i \le q. \tag{25}$$

The Frobenius first-Chern-class identity is also [13, Lemma 4.2]; it is an identity of rational divisor classes. For the second equality, use (18). Complementary degrees in (19) have total first Chern class $e_j(p-1)K_W$. Summing gives

$$\sum_j c_1(A_j(\Omega^1_W)) = \frac{s(p-1)}{2} K_W.$$

Thus $c_1(F^*\mathcal{E}) = sB_p + s(p-1)K_W/2$, and Frobenius pullback of twisted divisor classes multiplies by p, as required.

For $0 \le \lambda \le q$, the fixed movable inequality gives

$$(G + K_W/2 + \lambda S) \cdot \gamma \le 40\epsilon J \cdot \gamma,$$

where pullback by π_x is understood. This concerns only intersections of fixed divisors and the fixed complete-intersection witness, so it remains true on the reduction. The class in (25) differs from its left-hand divisor at $\lambda = \lambda_i$ by

$$\frac{1}{p}\big(B_p - pD - K_W/2\big).$$

Its numerator belongs to the fixed finite list $T_0 - aD - K_W/2$, $a \in \{0, \dots, k_0 - 1\}$. The resulting error after intersection with γ is therefore $O(p^{-1})$, uniformly over the selected determinant columns.

Let x' be the twisted point and let $0 \ne \tau_i \in H^0(W', \mathcal{Q}_i)$. The strict transform of its effective divisor on the blowup at x' pairs nonnegatively with the twist of γ. Since $J \cdot \gamma > 0$, it follows that

$$\frac{\operatorname{ord}_{x'}(\tau_i)}{R} \le 40\epsilon + O(1/p). \tag{26}$$

The constant is uniform in the columns and in τ_i. This argument tests sections on the reduction itself; it does not require them to lift to characteristic zero.

Künneth identifies the space containing σ with $H^0(W', \mathcal{Q}_1) \otimes H^0(W', \mathcal{Q}_2)$. Choose bases adapted to the orders of vanishing at x' in each factor. In each degree their leading terms

are linearly independent. Tensor products of these leading terms remain independent in each bidegree of the two disjoint sets of parameters; terms of different bidegrees cannot cancel. Every nonzero tensor consequently has order at most the sum of the two maximum basis orders. Applying (26), we obtain

$$\operatorname{ord}_{(x',x')} \sigma \le R\big(80\epsilon + O(1/p)\big). \tag{27}$$

On the other hand, (16) and Lemma 5.5 bound the matrix rank at (x', x') by a number strictly smaller than $R/4$. Trivialize its line bundles near that point. Invertible constant row operations make more than $3R/4$ rows vanish at the point. Each entry of those rows belongs to the maximal ideal, so every term of the determinant has order at least $3R/4$. Therefore

$$\operatorname{ord}_{(x',x')} \sigma \ge 3R/4. \tag{28}$$

Equations (27) and (28) contradict $80\epsilon < 3/4$ for sufficiently large p.

This contradiction proves Proposition 5.1. The full-rank calculation used z_*, while the diagonal estimate holds at every point and was tested at the separately chosen x. The errors were controlled by one fixed finite list of divisor classes and one fixed flag polynomial. Thus their constants do not depend on p, the filtration index, or the chosen columns.

6. Proof of Lemma 1.2

Proof. Fix $s > 0$. Choose n sufficiently large that Lemma 4.1 applies, $r_n > s + 1$, and $P_n^d < (2\epsilon)^d$. Since $P_n - r_n D$ is nef,

$$(2P_n + sD) \cdot P_n^{d-1} \le (2 + s/r_n) P_n^d < 4P_n^d.$$

Fix $x, \beta, E, f, H_1, \dots, H_{d-1}$ supplied by that lemma. For every $0 \le \lambda \le s$, nefness of D gives

$$f^*\beta^*(P_n + \lambda D) \cdot H_1 \cdots H_{d-1} < 40\epsilon\, f^*E \cdot H_1 \cdots H_{d-1}.$$

Choose the flag in Lemma 4.2 with $P = P_n$.

If $|kR_{n,s}|$ separated $(2d+2)k$-jets at any $z \in U$, these fixed data would satisfy all hypotheses of Proposition 5.1, with

$$W = X, \qquad G = H = P_n, \qquad S = D, \qquad q = s, \qquad r = r_n.$$

Here $K_X \sim 0$, so the canonical-intersection condition is automatic. This is impossible. Hence no such k, z exist. □

Remark. This obstruction uses (I_d) and $D^d = 0$, but does not use $\kappa(D) = -\infty$. That extra assumption retains the nonvanishing proof format of the threefold framework. Under the source inputs, the two jet lemmas actually exclude the entire full-nef-dimension, non-big case.

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